

Proof Techniques

Notes By Pr. El Hadiq Zouhair

12.1 The vocabulary of mathematical statements

A *theorem* is a statement proved true; a *lemma* is a helper theorem; a *corollary* follows easily from a theorem; a *conjecture* awaits proof; a *proof* is a valid argument, built from axioms, definitions and previous results by the rules of inference of Chapter 5. Most theorems have the form $\forall x (P(x) \rightarrow Q(x))$, and the techniques below are strategies for establishing $P \rightarrow Q$ for an arbitrary x (whence UG concludes).

12.2 Direct proof

To prove $p \rightarrow q$ directly: assume p , derive q by definitions and known results.

Theorem 12.1. If n is odd then n^2 is odd.

Proof. Let $n = 2k + 1$ for some integer k . Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. $\square$

12.3 Proof by contraposition

To prove $p \rightarrow q$, prove $\neg q \rightarrow \neg p$ instead — the two are equivalent (Chapter 2).

Theorem 12.2. If n^2 is even then n is even.

Proof. Contrapositive: if n is odd then n^2 is odd — proved above. $\square$

Choose contraposition when $\neg q$ is more concrete to work with than p (here: “ n odd” gives an explicit form, “ n^2 even” does not).

12.4 Proof by contradiction

To prove φ : assume $\neg\varphi$ and derive a contradiction ($r \wedge \neg r$); then $\neg\varphi$ is impossible, so φ holds.
Template: “Suppose, for contradiction, that ... Then ... contradiction. Hence ...”

Theorem 12.3 (Canonical example). $\sqrt{2}$ is irrational.

Proof. Suppose $\sqrt{2} = a/b$ with $a, b \in \mathbb{Z}^+$ in lowest terms. Then $a^2 = 2b^2$, so a^2 is even, so a is even (previous theorem): $a = 2c$. Then $4c^2 = 2b^2$, i.e. $b^2 = 2c^2$, so b is even too — contradicting lowest terms. Hence $\sqrt{2}$ is irrational. $\square$

Theorem 12.4 (Euclid). There are infinitely many primes.

Proof. Suppose $p_1, \dots, p_k$ were all the primes and set $N = p_1 \cdots p_k + 1$. Each p_i divides $N - 1$, so none divides N ; yet $N > 1$ has a prime divisor — a prime outside the list: contradiction. $\square$

12.5 Proof by cases

To prove φ , split the situation into finitely many exhaustive cases and prove φ in each — justified by $(p_1 \vee \cdots \vee p_n) \wedge \bigwedge_i (p_i \rightarrow q) \models q$.

Theorem 12.5. For every integer n , $n^2 \geq n$.

Proof. Case $n \leq 0$: then $n^2 \geq 0 \geq n$. Case $n \geq 1$: multiplying $n \geq 1$ by $n > 0$ gives $n^2 \geq n$. The cases are exhaustive. $\square$

12.6 Existence proofs

An existence statement $\exists x P(x)$ may be proved *constructively* — exhibit a witness — or *non-constructively* — show a witness must exist without naming one.

Example 12.6 (Constructive). There is a positive integer expressible as a sum of two cubes in two ways: $1729 = 1^3 + 12^3 = 9^3 + 10^3$.

Example 12.7 (Non-constructive). There exist irrationals x, y with x^y rational. Consider $z = \sqrt{2}^{\sqrt{2}}$. If z is rational, take $x = y = \sqrt{2}$. If z is irrational, take $x = z$, $y = \sqrt{2}$: then $x^y = \sqrt{2}^2 = 2$. Either way a witness exists — yet we have not determined which.

12.7 Uniqueness proofs

To prove $\exists! x P(x)$: prove existence, then prove uniqueness — assume $P(a)$ and $P(b)$ and derive $a = b$.

Example 12.8. The equation $ax = b$ ($a \neq 0$) has the unique solution $x = b/a$: existence by substitution; uniqueness since $ax_1 = b = ax_2$ implies $x_1 = x_2$ after dividing by $a \neq 0$.

12.8 Disproof by counterexample

To refute $\forall x P(x)$, one counterexample suffices.

Example 12.9. “Every positive integer is a sum of two squares” is false: 3 is not. “ $2^{2^n} + 1$ is prime for all n ” (Fermat) fails at $n = 5$: $2^{32} + 1 = 641 \times 6700417$ (Euler).

12.9 Common errors

Beware: proving $p \rightarrow q$ by assuming q (begging the question); arguing from examples (“it works for $n = 1, 2, 3$, hence always”); dividing by a quantity that may be zero; affirming the consequent; and the hidden assumption that objects named differently are distinct. A classic bogus proof that $1 = 2$ divides both sides of $a^2 - ab = a^2 - b^2$ by $a - b$ when $a = b$ — division by zero in disguise.

12.10 Strategy

Given a conjecture $p \rightarrow q$: try *direct* first; if $\neg q$ unpacks better, try *contraposition*; if both directions stall or the statement is a pure negative (“is irrational”, “no largest ...”), try *contradiction*; if natural subcases appear (parity, sign, remainders), use *cases*. For $\exists$, look for an explicit witness before resorting to non-constructive arguments. And before proving anything hard, hunt briefly for a counterexample — failure to find one often reveals the proof idea. Statements quantified over all $n \in \mathbb{N}$ usually call for *induction*, the subject of Chapter 13.

12.11 Summary

Direct, contrapositive, contradiction and cases prove implications; witnesses or indirect arguments prove existence; existence plus “two solutions are equal” proves uniqueness; one counterexample refutes a universal claim. The two jewels — irrationality of $\sqrt{2}$ (Theorem 12.3) and Euclid’s infinitude of primes (Theorem 12.4) — model the contradiction template every mathematician carries.

Exercises

Exercise 12.1 EASY

Prove directly: the sum of two odd integers is even.

Exercise 12.2 EASY

Prove directly: if $a \mid b$ and $b \mid c$ then $a \mid c$.

Exercise 12.3 EASY

Find counterexamples: (a) “every prime is odd”; (b) “if $n^2 > 0$ then $n > 0$ ”; (c) “the sum of two irrationals is irrational”.

Exercise 12.4 MEDIUM

Prove by contraposition: if $3n + 2$ is odd, then n is odd.

Exercise 12.5 MEDIUM

Prove by contradiction: there is no smallest positive rational number.

Exercise 12.6 MEDIUM

Prove by cases: for all integers n , $3 \mid n(n+1)(n+2)$.

Exercise 12.7 MEDIUM

Prove that $\sqrt{3}$ is irrational, adapting the proof of Theorem 12.3. Where would the same argument break for $\sqrt{4}$?

Exercise 12.8 MEDIUM

Prove there exists a unique real x with $x^3 + x - 2 = 0$. (*Existence: exhibit it. Uniqueness: factor or use monotonicity.*)

Exercise 12.9 HARD

Prove that if n is a positive integer and n^2 is divisible by 3, then n is divisible by 3 (use cases on $n \bmod 3$); this is the lemma your $\sqrt{3}$ proof needed.

Exercise 12.10 HARD

Show non-constructively that in any group of 2 or more people at a party, two people have shaken the same number of hands (handshakes within the group). *Hint: pigeonhole on the possible counts.*

Exercise 12.11 HARD

Criticise this “proof” that every positive integer n equals the next: “ $n = n + 1$ implies $(n)^2 - (n + 1)n = n^2 - (n + 1)n$, both sides simplify to $-n = -n$, true; hence $n = n + 1$.” Identify the logical error precisely.

Solutions to the Exercises

Solution 12.1. Let $m = 2j + 1$, $n = 2k + 1$. Then $m + n = 2(j + k + 1)$, an even number. ■

Solution 12.2. $b = aj$ and $c = bk$ for integers j, k ; then $c = a(jk)$, so $a \mid c$. ■

Solution 12.3. (a) 2. (b) $n = -1$: $n^2 = 1 > 0$ but $n < 0$. (c) $\sqrt{2} + (-\sqrt{2}) = 0 \in \mathbb{Q}$. ■

Solution 12.4. Contrapositive: if n is even, $3n+2$ is even. Let $n = 2k$: $3n+2 = 6k+2 = 2(3k+1)$, even. Done. ■

Solution 12.5. Suppose $r > 0$ were the smallest positive rational. Then $r/2$ is rational, positive, and smaller — contradiction. ■

Solution 12.6. Among any three consecutive integers one is divisible by 3: by cases on $n \bmod 3$. If $n \equiv 0$, then $3 \mid n$; if $n \equiv 1$, then $3 \mid n + 2$; if $n \equiv 2$, then $3 \mid n + 1$. In each case the product is divisible by 3. ■

Solution 12.7. Suppose $\sqrt{3} = a/b$ in lowest terms; then $a^2 = 3b^2$, so $3 \mid a^2$, hence $3 \mid a$ (Exercise 12.9): $a = 3c$, so $3b^2 = 9c^2$, $b^2 = 3c^2$, and $3 \mid b$ — contradicting lowest terms. For $\sqrt{4}$, the lemma fails: $4 \mid a^2$ does *not* force $4 \mid a$ (take $a = 2$), and indeed $\sqrt{4} = 2$ is rational. ■

Solution 12.8. Existence: $x = 1$ works: $1 + 1 - 2 = 0$. Uniqueness: $x^3 + x - 2 = (x - 1)(x^2 + x + 2)$, and $x^2 + x + 2 = (x + \frac{1}{2})^2 + \frac{7}{4} > 0$ has no real roots; alternatively $f(x) = x^3 + x - 2$ is strictly increasing ($f'(x) = 3x^2 + 1 > 0$), so it crosses zero exactly once. ■

Solution 12.9. Cases. If $n = 3k$: done, $3 \mid n$. If $n = 3k + 1$: $n^2 = 9k^2 + 6k + 1 \equiv 1 \pmod{3}$, not divisible. If $n = 3k + 2$: $n^2 = 9k^2 + 12k + 4 \equiv 1 \pmod{3}$, not divisible. So $3 \mid n^2$ occurs only in the first case, where $3 \mid n$. ■

Solution 12.10. With n people, each handshake count lies in $\{0, 1, \dots, n - 1\}$ — n values. But counts 0 and $n - 1$ cannot both occur (someone who shook everyone's hand rules out someone who shook none). So at most $n - 1$ values are available for n people: by pigeonhole two people share a count. No specific pair is exhibited — the proof is non-constructive. ■

Solution 12.11. The argument assumes $n = n + 1$ *first* and derives a true statement. Deriving a truth from an assumption proves nothing about the assumption (that is affirming the consequent at the level of proofs); a valid proof must derive the *conclusion* from accepted premises, not the reverse. The implication chain is also reversible only if every step is an equivalence, which squaring-style manipulations are not. ■